

Project Part 1 Presentation Script (10 minutes)

Shelter Occupancy / Capacity (City of Toronto)

Mani

Yanness

Tyler

Even split plan (3 speakers)

We split the talk into **three equal sections** of about **3 minutes 20 seconds each**.

- **Mani:** Slides 1–2 (Intro + Motivation + Hypothesis) (0:00–3:20)
- **Yanness:** Slides 3–5 (Dataset + Preprocessing + Ethics) (3:20–6:40)
- **Tyler:** Slides 6–8 (Algorithms + Visualizations + Conclusion/Q&A) (6:40–10:00)

Timing overview

Time	Slide(s)	Speaker	Goal
0:00–3:20	1–2	Mani	Introduce team/topic, explain motivation, state hypothesis
3:20–6:40	3–5	Yanness	Present dataset URLs/fields, preprocessing plan, ethics/limitations
6:40–10:00	6–8	Tyler	Justify algorithms (FFT/QR/LU/clustering), visuals, next steps, Q&A

Slide-by-slide script

Slide 1 — Title & Team (0:00–0:45)

Speaker: Mani

Script (Mani):

Hi everyone. We are **Mani, Yanness, and Tyler**. Our project is about **shelter occupancy and capacity in Toronto**.

We will use daily shelter and overnight service data to understand **system pressure** and **patterns over time**.

Today we are presenting our plan: our goal, dataset, preprocessing, ethics, algorithms, and planned visualizations.

Slide 2 — Motivation & Hypothesis (0:45–3:20)

Speaker: Mani

Script (Mani):

This topic matters because shelter spaces are limited, and demand can change over time.

If we can find regular patterns and identify high-pressure programs, the analysis can support planning and resource decisions.

Hypothesis (H1): Occupancy rate shows **weekly seasonality**, and some programs are **persistently high pressure**.

We also expect pressure to differ by **program/service type** and by **capacity type**, like **actual capacity vs funding capacity**.

Our measurable outputs are occupancy rates, pressure thresholds like “ $\geq 95\%$ occupancy”, and seasonality strength.

Slide 3 — Dataset Overview (3:20–4:30)

Speaker: Yanness

Script (Yanness):

Our main dataset is the **Daily Shelter & Overnight Service Occupancy & Capacity** dataset from the City of Toronto.

It contains daily records at the program level, including occupancy and several capacity measures.

We will use the catalogue page and direct CSV export links for downloading and documentation.

We also note it is a daily snapshot and updates may not be posted on weekends or holidays, so we will account for that.

Slide 4 — Preprocessing Plan (4:30–5:45)

Speaker: Yanness

Script (Yanness):

For preprocessing, we will parse dates and select a consistent time window, like 2021 to 2024.

We will clean missing values and handle edge cases like capacity equal to zero.

We will compute key metrics:

$$\text{occupancy_rate_actual} = \frac{\text{occupied}}{\text{actual_capacity}} \quad \text{occupancy_rate_funding} = \frac{\text{occupied}}{\text{funding_capacity}}$$

We will also add time features such as day-of-week and month, and create pressure indicators like the percent of days above 95% occupancy.

Finally, we will aggregate results at the system level and by program/service categories.

Slide 5 — Ethics & Bias / Limitations (5:45–6:40)

Speaker: Yanness

Script (Yanness):

This topic is sensitive, so we will communicate results carefully and avoid stigmatizing specific locations or groups.

We will focus on aggregate patterns and clearly state limitations.

Reporting gaps on weekends or holidays can affect daily trends, so we will check for these issues.

Also, capacity definitions can differ, so we will document whether we use actual or funding capacity in each analysis.

Slide 6 — Algorithms (+ QR / LU / FFT) (6:40–8:20)

Speaker: Tyler

Script (Tyler):

We have three analysis parts.

First, FFT for seasonality. We apply the Fast Fourier Transform to the occupancy-rate time series to detect periodic cycles, like weekly patterns, and measure how strong they are.

Second, regression for explanation and prediction. We build a baseline regression model that predicts next-day occupancy rate using time features and lag features. This gives interpretable coefficients and a simple forecasting baseline.

Third, clustering for pressure profiles. For each program, we create features like mean occupancy, variance, and percent of days above 95% occupancy, and then cluster programs into a few pressure categories.

To connect to course methods, we can solve least squares using **QR factorization** for numerical stability, and use **LU decomposition** for efficient linear system solving.

Slide 7 — Visualization Plan (8:20–9:20)

Speaker: Tyler

Script (Tyler):

We will use visualizations to communicate patterns clearly.

We will make a line chart of occupancy rate over time at the system level.

We will use a heatmap of day-of-week by month to show seasonality.

We will make a bar chart showing percent of days each program is above 95% occupancy as a pressure indicator.

And we will show cluster results with a scatter plot or profile chart. If we add a locations dataset, we can also make a map view later.

Slide 8 — Conclusion & Next Steps (9:20–10:00)

Speaker: Tyler

Script (Tyler):

To summarize, we will analyze Toronto's shelter occupancy and capacity to measure pressure, detect seasonality, and group programs by usage patterns.

Next steps are: download and clean the data, compute occupancy rates, run FFT, regression, and clustering, and then create the planned visualizations.

Thank you — we welcome any questions or feedback.